

Midterm Practice

- Use the questions in this question bank to practice for the midterm test.
- **Syllabus:** everything covered up to and including the week before the midterm test.

Reminders:

- The midterm counts for **20%** of your final grade.
- Both the midterm and the final test are **closed book**.
 - However, you are allowed to bring 1 A4 size sheet.
 - You are also allowed to bring your calculator.

Pandas Series Questions:

1. Which method is used to add values in a Pandas Series?
(a) add()
(b) sum()
(c) plus()
(d) append()
2. How do you perform element-wise multiplication in a Pandas Series?
(a) multiply()
(b) product()
(c) mult()
(d) *
3. Which of the following is used to handle missing values during arithmetic operations in Series?
(a) fill_value
(b) na_fill
(c) fillna
(d) add_fill
4. What does the divide() function do in a Pandas Series?
(a) Divides each element by a single number
(b) Adds each element by the corresponding element of another array
(c) Divides each element by the corresponding element of another array
(d) Subtracts each element by a single number
5. What is the default fill_value in a Pandas add operation?
(a) 0
(b) 1
(c) NaN
(d) None
6. When using the add() method, which parameter allows handling NaN values?
(a) missing
(b) nan_replace
(c) fill_value
(d) nan_fill

7. Which of these will multiply each element of Series 's' by 10?
- (a) `s * 10`
 - (b) `s.add(10)`
 - (c) `s + 10`
 - (d) `s.multiply(10.0)`
8. Which code will add Series 'a' and 'b', filling NaNs in 'a' with 0?
- (a) `a.add(b, fill_value=0)`
 - (b) `a.add(b, fill=0)`
 - (c) `a + b`
 - (d) `add(a, b, 0)`
9. How does the `add()` method differ from `+` in Series?
- (a) It cannot handle NaN
 - (b) It allows `fill_value`
 - (c) It is faster
 - (d) It does element-wise addition
10. What does `s.divide(2)` do in Series 's'?
- (a) Divides all by 2
 - (b) Divides NaN by 2
 - (c) Returns None
 - (d) Divides first element by 2

Pandas DataFrames Questions:

1. How can you load a CSV file into a DataFrame?
- (a) `pd.read_csv()`
 - (b) `pd.load_csv()`
 - (c) `pd.to_csv()`
 - (d) `pd.open_csv()`
2. Which method selects a single row?
- (a) `loc[]`
 - (b) `select[]`
 - (c) `get[]`
 - (d) `row[]`
3. Which function removes columns in a DataFrame?
- (a) `drop()`
 - (b) `delete()`
 - (c) `remove()`
 - (d) `rm()`
4. How can you import datasets from sklearn?
- (a) `sklearn.datasets.load()`
 - (b) `sklearn.data()`
 - (c) `load.sklearn()`
 - (d) `sklearn.dataset()`

5. If my_DF is a Pandas DataFrame with columns labeled 'c1' and 'c2', what will the operation

my_DF['c3'] = 100.0 * my_DF['c1']/my_DF['c2']

result in?

- (a) Runtime error. Series are immutable
- (b) Creating a new column labeled 'c3'
- (c) Runtime error. That is not how elementwise division is performed with DataFrame columns.
- (d) Creating a new column labeled 'c3' but an 'inplace' command is required to permanently add the column to my_DF

6. Which method selects multiple columns by labels?

- (a) loc[:, [cols]]
- (b) loc[:, col]
- (c) iloc[col]
- (d) col[]

7. How can you drop multiple columns by name?

- (a) df.drop(columns=[...])
- (b) drop.multicol(...)
- (c) delete.cols([...])
- (d) rm([...])

8. How do you append a new DataFrame row with values [1, 2, 3]?

- (a) df.loc[len(df)] = [1,2,3]
- (b) df.append(1,2,3)
- (c) df.new_row([1,2,3])
- (d) add.row([...])

9. Which method allows selecting rows by index numbers?

- (a) iloc[]
- (b) loc[]
- (c) select[]
- (d) getrow()

10. How can you adjust all DataFrame columns by multiplying by 2?

- (a) df * 2
- (b) df.mul(2)
- (c) both
- (d) neither

Conditional Filtering Questions:

1. Which method filters rows based on a list of values?

- (a) isin()
- (b) inlist()
- (c) contains()
- (d) filter()

2. How do you filter rows within a certain range of values?

- (a) `between()`
- (b) `range()`
- (c) `within()`
- (d) `slice()`

3. Which operator is used for logical AND in Pandas?

- (a) `&`
- (b) `and`
- (c) `&&`
- (d) `AND`

4. To delete rows that match a condition, you can use:

- (a) `drop()`
- (b) `delete_rows()`
- (c) `remove_rows()`
- (d) `del()`

5. How do you filter rows based on a single condition in Pandas?

- (a) `df[condition]`
- (b) `df.where()`
- (c) `df.filter()`
- (d) `df.query()`

6. Which of these is a valid example of conditional filtering based on multiple conditions?

- (a) `df[(cond1) & (cond2)]`
- (b) `df{cond1 && cond2}`
- (c) `filter([cond1, cond2])`
- (d) `df[cond1, cond2]`

7. Which is true about the `isin()` method?

- (a) It checks if values exist
- (b) It can filter columns
- (c) It replaces values
- (d) It drops values

8. Which of these can be used to drop rows between two values using conditional filtering?

- (a) `df[~df[col].between(val1, val2)]`
- (b) `df.drop(val1, val2)`
- (c) `df[val1:val2]`
- (d) `remove(df.between)`

9. What would `df[df['col'].isin([1, 2])]` return?

- (a) Rows with `col=1` or `col=2`
- (b) Rows with `col` not 1 and 2
- (c) Rows with `col` only 1
- (d) All rows

10. To filter rows outside a range, use:

- (a) `~df[col].between(val1, val2)`
- (b) `df.drop([val1, val2])`
- (c) `df[not val]`
- (d) `reverse_between(val)`

Useful Pandas Methods Questions:

1. Which method provides summary statistics of a DataFrame?

- (a) `stats()`
- (b) `info()`
- (c) `describe()`
- (d) `stats_summary()`

2. How do you count unique values in a column?

- (a) `nunique()`
- (b) `unique()`
- (c) `countunique()`
- (d) `unique_count()`

3. To remove duplicate rows in DataFrame:

- (a) `unique()`
- (b) `drop_duplicates()`
- (c) `delete_duplicates()`
- (d) `drop()`

4. Which method returns column labels sorted by values?

- (a) `sort_values()`
- (b) `order_by()`
- (c) `sort()`
- (d) `arrange()`

5. Which method returns the index of the minimum value in a column?

- (a) `locate_min()`
- (b) `idx_min()`
- (c) `index_min()`
- (d) `idxmin()`

6. Which method applies a function to each row or column?

- (a) `apply()`
- (b) `execute()`
- (c) `transform()`
- (d) `map()`

7. What is the result of `value_counts(normalize=True)`?

- (a) Proportion of each unique value
- (b) Total count

- (c) Sum of values
- (d) None

8. Which code would sort 'df' by 'col1' in ascending order?

- (a) `df.sort_values(by='col1')`
- (b) `df.sort('col1')`
- (c) `order(df, 'col1')`
- (d) `sort()`

9. How does `df['col'].map(lambda x: x**2)` work?

- (a) Multiplies 'col' values by 2
- (b) Changes 'col' data type
- (c) Raises 'col' values to the power of 2
- (d) None of the above

10. What does `df.corr()` return?

- (a) Correlation matrix across columns
- (b) Column counts
- (c) Duplicate rows
- (d) Correlation matrix across rows

Pandas Series Answers:

- 1. (a)
- 2. (a)
- 3. (a)
- 4. (b), (c)
- 5. (c)
- 6. (c)
- 7. (a), (d)
- 8. (a)
- 9. (b)
- 10. (a)

Pandas DataFrames Answers:

- 1. (a)
- 2. (a)
- 3. (a)
- 4. (a)
- 5. (b)

6. (a)

7. (a)

8. (a)

9. (a)

10. (c)

Conditional Filtering Answers:

1. (a)

2. (a)

3. (a)

4. (a)

5. (a)

6. (a)

7. (a)

8. (a)

9. (a)

10. (a)

Useful Pandas Methods Answers:

1. (c)

2. (a)

3. (b)

4. (a)

5. (d)

6. (a)

7. (a)

8. (a)

9. (c)

10. (a)

